

Mahmoud Mohamed Attia

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Career Objective

Computer Engineering undergraduate with a strong foundation in Data Structures, Algorithms, and Scalable Backend Systems. Experienced in architecting complex applications using NestJS, Docker, and Microservices. Proven ability to integrate AI models (Gemini) and Real-time WebSockets into production-ready environments. Seeking to leverage deep technical expertise in the Noon CAKE Program to build high-performance e-commerce solutions.

Education

Cairo University, Faculty of Engineering | Cairo, Egypt *Bachelor of Science in Computer Engineering* | Expected June 2027

Relevant Coursework: Advanced Data Structures, Analysis of Algorithms, Operating Systems, Database Management Systems, Distributed Systems, Computer Networks.

Awards: 3rd Place, TCCD 13th Research Competition (Statistical Modeling & Data Analysis).

Certifications

- CompTIA A+ (Online, 2025)
- Cisco Certified Network Associate (CCNA) (Online, 2025)
- Introduction to Robotics (OnCampus, 2024)
- TCCD Research Event 3rd place (2025)

Technical Skills

- **Languages:** C++, Java, TypeScript, JavaScript, Python, C#, SQL.
- **Backend & Cloud:** Node.js, NestJS, Express, Docker, Google Cloud Platform (GCP), REST APIs, Microservices Architecture.
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS, Vite.
- **Databases:** PostgreSQL (Complex Relations), MongoDB (Aggregation Pipelines), Redis.
- **Concepts:** OOP, SOLID Principles, System Design, CI/CD, Monorepo Management (Turborepo).

Professional Experience

Software Development

Full Stack Developer Intern (Remote) | Codveda [Aug 2025] – [Oct2025]

- Developed and maintained responsive web applications using **React.js** and **Node.js**, ensuring cross-browser compatibility.
- Integrated third-party RESTful APIs and optimized backend endpoints to improve data retrieval latency.
- Collaborated in an **Agile/Scrum** environment, using Git for version control and participating in code reviews to maintain code quality.

IT Infrastructure Intern | iScore (The Egyptian Credit Bureau) [Jul 2025] – [Aug2025]

- Gained hands-on exposure to enterprise-grade network infrastructure and server management in a high-security financial environment.

- Assisted in monitoring network uptime and troubleshooting connectivity issues, ensuring adherence to strict SLAs.
- Developed the iScore mobile app using React native.js & node.js.

Featured Projects

MedConnect (Medical Referral Enterprise System) | *NestJS, PostgreSQL, Docker, Turborepo* Jan 2026
[GitHub](#)

- Architected a **Monorepo-based** B2B2C platform connecting doctors and clinics, designed for high scalability.
- Engineered a secure backend using **NestJS** and **Prisma ORM**, implementing complex **Many-to-Many relationships** for patient referrals.
- Deployed **Dockerized** containers for the database and API services to ensure environment consistency.
- Implemented **Role-Based Access Control (RBAC)** using JWT and Bcrypt guards to secure sensitive medical data.

Real-Time Collaborative Text Editor

May 2025

CMPS211 Course Project, Cairo University • [GitHub](#)

- Built a real-time text editor in Java using CRDTs for multi-user editing.
- Enabled live updates and cursor tracking for 10+ simultaneous users.
- Led team of four, delivering documented application on time.

Focus Ritual: Productivity Web Application

April 2025

Personal Project • [GitHub](#)

- Developed task management web app using React and Node.js with RESTful APIs.
- Optimized frontend for 20% faster load times with HTTPS deployment.
- Documented scalable codebase for future enhancements.

Network & Security

Technical Support Specialist, Smart Electronics

June 2021 – May 2025

Cairo, Egypt

- Resolved 50+ weekly hardware/network issues, boosting customer satisfaction.
- Configured routers and switches for client system compatibility.
- Applied CompTIA A+ skills in PC assembly and diagnostics.

Network Simulation Project

March 2025

CCNA Course Project

- Configured enterprise network in Packet Tracer with VLANs and OSPF.
- Optimized performance using Wireshark, achieving 95% uptime.
- Documented and presented configurations to peers.

Hardware & Systems

Arithmetic Logic Unit (ALU) Design

November 2024

Digital Logic Design Course Project, Cairo University

- Designed 4-bit ALU in Verilog for arithmetic and logic operations.
- Tested on FPGA with 95% test case success rate.
- Documented power and performance optimizations.

Volunteer Experience

Volunteer Developer, Enactus Egypt
Cairo, Egypt

December 2024 – Present

- Developed accessibility tools for individuals with disabilities.
- Supported network infrastructure for underserved communities.
- Collaborated on mobile-first projects in cross-functional teams.

Achievements

- Secured 3rd place in TCCD 13th Edition Research Competition, analyzing failure reports to identify patterns, distinguish mechanical vs. human factors, build predictive statistical models, and visualize data for aviation safety protocols (Cairo, Egypt, May 2025).
- Volunteered with Enactus Cairo University, contributing to entrepreneurial projects for social impact (2024).
- Shared CCNA study notes on LinkedIn using Notion and Canva, boosting profile engagement (2025).
- Solved algorithmic challenges in Cairo University Coding Competition using Java (2024).

Languages

- Arabic (Native)
- English (Fluent)